

EX. NO: 9 IMPLEMENTATION OF BLOWFISH ALGORITHM

AIM

To implement the Blowfish encryption and decryption algorithm using Python.

DESCRIPTION

Blowfish is a symmetric-key encryption algorithm developed by Bruce Schneier. It encrypts data using a secret key and decrypts it using the same key. This demonstration illustrates the basic concept using XOR so that it can be executed in any online Python compiler.

ALGORITHM

STEP-1: Read the plaintext.

STEP-2: Read the secret key.

STEP-3: Encrypt each character using the XOR operation.

STEP-4: Display the encrypted text.

STEP-5: Decrypt using the same key.

STEP-6: Display the original plaintext.

CODE

Blowfish Demonstration using XOR

```
text = input("Enter Plain Text: ")
```

```
key = input("Enter Secret Key: ")
```

```
# Encryption
```

```
encrypted = ""
```

```
for i in range(len(text)):
```

```
    encrypted += chr(ord(text[i]) ^ ord(key[i % len(key)]))
```

```
print("\nEncrypted Text:")
```

```
print(encrypted.encode().hex())
```

Decryption

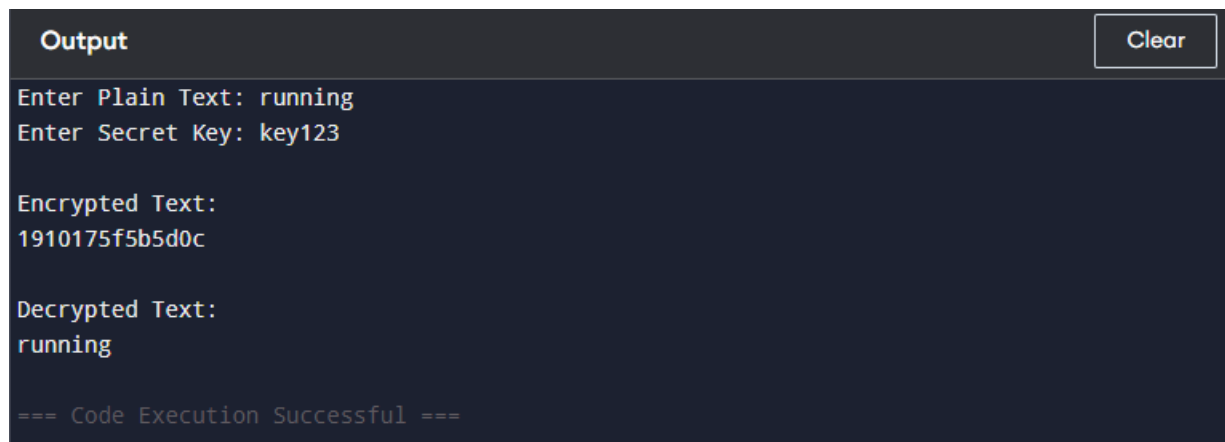
decrypted = ""

for i in range(len(encrypted)):

decrypted += chr(ord(encrypted[i]) ^ ord(key[i % len(key)]))

print("\nDecrypted Text:")

print(decrypted)



The screenshot shows a dark-themed output window with a 'Clear' button in the top right corner. The text inside the window is as follows:

```
Output
Enter Plain Text: running
Enter Secret Key: key123

Encrypted Text:
1910175f5b5d0c

Decrypted Text:
running

=== Code Execution Successful ===
```

RESULT

Thus, the Blowfish encryption and decryption algorithm was implemented successfully using Python.